

Smart Hoax Detector: Benchmarking NLP Models for Misinformation Political Classification in Indonesia

1st Brian Sanjaya Halim
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
brian.halim@binus.ac.id

2nd Radianda Setiawan
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
radianda.setiawan@binus.ac.id

3rd Albertus Tejasukmana
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
albertus.tejasukmana@binus.ac.id

4th Jovan Immanuel Manik
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
jovan.manik@binus.ac.id

Abstract—The rapid proliferation of digital information has led to a significant increase in the dissemination of fake news and hoaxes, posing a critical threat to public trust and social stability. While various Natural Language Processing (NLP) models have been developed for text classification, accurately detecting hoaxes within localized, non English contexts such as the Indonesian digital landscape presents unique linguistic challenges. This study engineers a comparative evaluation framework to benchmark five distinct architectures for Indonesian hoax detection, ranging from classical machine learning to advanced deep learning models: a Support Vector Machine (SVM) with TF-IDF vectorization, a 1D Convolutional Neural Network (1D-CNN), a Bidirectional Long Short-Term Memory network with an Attention mechanism (BiLSTM+Attention), IndoBERT V2, and the multilingual XLM-RoBERTa. Empirical evaluation demonstrated that transformer-based architectures exhibit a definitive superiority in capturing deep semantic structures. IndoBERT V2 achieved the peak performance with a macro F1-score of 99.39%, closely followed by XLM-RoBERTa at 99.12%, while all deployed models maintained a robust performance baseline between 98.00% and 99.39%. Furthermore, qualitative validation through user acceptance testing (UAT) confirmed that the exceptionally high classification accuracy is strongly attributed to the models' capacity to extract stylistic features. The neural networks successfully differentiated the standardized, formal register of professional journalism from the informal, colloquial phrasing characteristic of user-generated misinformation. These findings conclude that domain-specific pre-trained transformers offer unparalleled efficacy for automated Indonesian hoax detection, establishing a highly reliable framework for real world deployment.

Keywords—*Natural Language Processing, Hoax Detection, Text Categorization, Transformers, Stylometry, IndoBERT, XLM-RoBERTa, Indonesian NLP*

I. INTRODUCTION

The exponential growth of internet penetration and digital platform accessibility in Indonesia has fundamentally transformed information consumption patterns. However, this shift has concurrently accelerated the rapid propagation of digital misinformation and hoaxes, posing severe threats to public trust, democratic integrity, and social stability. Consequently, automated hoax detection has emerged as an imperative field of study within computational linguistics and Natural Language Processing (NLP) to build robust systems capable of mitigating the spread of harmful misinformation at scale [10].

Early computational approaches to text categorization primarily relied on statistical machine learning baselines. Traditional frameworks utilized Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction paired with Support Vector Machines (SVM) to construct high dimensional decision boundaries [6]. While these models remain exceptionally efficient and highly interpretable for keyword dense text categorization, they exhibit structural limitations in capturing sequential dependencies and long range semantic contexts within complex narratives [8], [9]. To overcome these limitations, the deep learning paradigm introduced neural networks capable of automatic feature engineering. One dimensional Convolutional Neural Networks (1D-CNN) were successfully deployed to capture local spatial n-gram patterns [4], while recurrent architectures, specifically Bidirectional Long Short Term Memory (BiLSTM) networks equipped with attention mechanisms, were engineered to map temporal and context heavy sentence dynamics [5], [7].

The contemporary state of the art in NLP is dominated by pre-trained bidirectional transformer architectures that leverage self attention mechanisms to model contextual token relations regardless of their distance [1]. While global cross lingual transformers like XLM-RoBERTa offer immense zero shot cross lingual capabilities [3], language-localized models such as IndoBERT V2 provide superior lexical alignment due to extensive pretraining on native Indonesian corpora [2]. Despite these advancements, a critical operational gap persists in evaluating how these diverse architectural families perform when subjected to identical localized constraints. Most notably, standard evaluations often overlook the underlying computational reasons for high model performance, specifically the capacity of deep neural networks to isolate stylometric feature patterns.

To address these challenges, this study presents a comprehensive comparative evaluation framework benchmarking five distinct computational paradigms: a linear SVM, a 1D-CNN, an attention based BiLSTM, IndoBERT V2, and XLM-RoBERTa for Indonesian hoax classification. The empirical findings demonstrate a definitive evolutionary progression, with transformer-based models achieving unparalleled predictive robustness. IndoBERT V2 established the peak performance benchmark with a macro F1-score of 99.39%, followed closely by XLM-RoBERTa at 99.12%. Furthermore, via user acceptance testing (UAT), this study reveals that the exceptional performance ceiling is deeply rooted in the models' capacity to successfully decode stylometric variations, systematically differentiating the formal register characteristic of mainstream journalism from the colloquial, high emotion syntax common in user generated hoaxes.

The remainder of this paper is structured systematically as follows: Section II reviews related literature and existing benchmarks; Section III details the data preprocessing pipeline and model configurations; Section IV presents the empirical results and evaluation analysis; and Section V concludes the study with directions for future work.

II. RELATED WORKS

The evolution of automated hoax and fake news detection has transitioned from reliance on traditional linguistic lexicons and statistical models to the deployment of advanced, context aware deep learning architectures. This section reviews the foundational methodologies and the recent paradigm shifts that inform the comparative framework of this study.

A. Traditional Machine Learning in Text Categorization

Early approaches to misinformation detection heavily utilized statistical feature extraction techniques paired with classical machine learning algorithms. Joachims [6] established that Support Vector Machines (SVM) are inherently well-suited for text categorization due to their

ability to handle high-dimensional, sparse feature spaces gracefully. In the context of Indonesian hoax detection, researchers have frequently employed TF-IDF vectorization combined with SVM to isolate recurring, formulaic deceptive keywords [8], [9]. While these models establish robust, lightweight, and interpretable baselines, they operate primarily on a "bag of words" assumption, structurally failing to preserve sentence syntax, word order, and deeper semantic nuances [10].

B. Deep Learning and Sequential Architectures

To address the contextual limitations of statistical baselines, the field shifted toward deep neural networks. Kim [4] demonstrated that 1D Convolutional Neural Networks (1D-CNN), originally designed for spatial image recognition, could be effectively adapted for sentence classification by capturing local n-gram patterns and short range textual dependencies. Concurrently, Recurrent Neural Networks (RNNs) were adopted to model temporal sequences in text. Specifically, Bidirectional Long Short Term Memory (BiLSTM) networks, especially when augmented with Attention mechanisms, proved highly effective in capturing both past and future contexts within complex sentences, allowing the model to dynamically weigh the importance of specific words regardless of their sequential distance [5], [7].

C. The Era of Pre Trained Transformers

The introduction of the self attention mechanism fundamentally disrupted NLP, culminating in the development of bidirectional transformers like BERT [1]. Transformers eliminated the sequential bottleneck of RNNs, enabling massive parallel processing and profound contextual understanding. For non English domains, Conneau et al. [3] introduced XLM-RoBERTa, a powerful cross-lingual language model pre-trained on 100 different languages, providing immense zero shot capabilities for localized misinformation detection.

However, applying global models to highly specific linguistic ecosystems often results in suboptimal lexical alignment. Addressing the unique syntax and morphology of the Indonesian language, Koto et al. [2] developed IndoBERT, a model rigorously pre-trained on diverse, localized Indonesian corpora. Existing literature strongly suggests that domain specific pretraining yields superior downstream performance in regional tasks [2]. This study builds directly upon these foundations by conducting a rigorous ablation study, directly benchmarking the traditional SVM, 1D-CNN, and BiLSTM architectures against the state of the art XLM-RoBERTa and IndoBERT V2 specifically within the Indonesian hoax detection paradigm.

III. METHODOLOGY

This section outlines the comprehensive architectural framework, data acquisition strategies, and experimental configurations formulated to conduct the ablation study for Indonesian hoax detection. To ensure strict empirical fairness and robust comparative analysis, all computational models were evaluated within an identical, isolated validation ecosystem.

A. Data Acquisition and Corpus Composition

The fundamental integrity of any natural language processing task relies heavily on the quality and diversity of its training corpus. For this study, the primary dataset was acquired from a publicly available repository on Kaggle, specifically the "Indonesian Fact and Hoax Political News" collection authored by M. Razif Rizqullah, Radhinansyah Hensa Ghaida, and Steve Immanuel Hadi.

To address the inherent class imbalance between the massive volume of factual journalism and the targeted hoax dataset, a rigorous downsampling strategy was implemented. The majority class ('FAKTA') was randomly sampled to exactly match the total volume of the minority class ('HOAKS'). This forced class parity (a strict 50:50 ratio) structurally ensures that the downstream predictive architectures do not develop an algorithmic bias toward the majority class prior to the 80/20 train-test split.

To construct a balanced and highly representative binary classification environment, the corpus was amalgamated from four distinct, large scale textual sources:

1. *dataset_cnn_10k*: Comprising approximately 10,000 articles scraped from CNN Indonesia.
2. *dataset_kompas_4k*: Containing roughly 4,000 articles sourced from Kompas.com.
3. *dataset_tempo_6k*: Consisting of 6,000 articles extracted from Tempo.co.
4. *dataset_turnbackhoax_10k*: Aggregating 10,000 debunked misinformation threads and user generated hoax samples from the TurnBackHoax.id database.

The articles derived from CNN, Kompas, and Tempo structurally represent the 'FAKTA' (Fact) class, characterized by formal journalistic registers and adherence to standardized Indonesian grammar (PUEBI). Conversely, the texts from TurnBackHoax represent the 'HOAKS' (Hoax) class, capturing the colloquial, often provocative stylometry of user generated misinformation. Prior to feature extraction, the aggregated corpus was deterministically partitioned into an 80% training subset and a 20% testing subset to strictly prohibit data leakage during model evaluation.

B. Textual Noise Reduction and Feature Engineering

Raw digital text is inherently unstructured and heavily corrupted by nonsemantic noise. To prepare the payloads for algorithmic ingestion, a rigorous multistage standardization pipeline was executed. Regular expression (Regex) filters were initially deployed to strip the text of HTML artifacts, uniform resource locators (URLs), alphanumeric anomalies, and non ASCII characters.

Following noise reduction, tokenization and feature engineering were bifurcated into two parallel execution paths to accommodate the distinct mathematical requirements of the deployed models:

1. *Vector Space Modeling (Statistical Path)*: For classical machine learning evaluation, the text was processed using Term Frequency Inverse Document Frequency (TF-IDF). To ensure the model captures sequential phrase contexts rather than isolated terms, the vectorizer was configured to extract both unigrams and bigrams (`ngram_range=(1, 2)`), limited to a maximum of 20,000 structural features. This technique maps clean token sequences into high-dimensional sparse matrices, mathematically penalizing ubiquitous words while amplifying the weight of rare, discriminative phrases indicative of deceptive intent.
2. *Sub-word Tokenization (Neural Path)*: For the deep learning and transformer architectures, payloads were serialized using advanced sub-word tokenization algorithms natively embedded within the Hugging Face ecosystem (e.g., WordPiece for BERT variants). To optimize tensor matrix operations within the neural networks, all tokenized sequences were statically padded or truncated to a uniform maximum sequence length of 128 tokens.

C. Architectural Frameworks for Ablation Study

To empirically benchmark the evolutionary progression of NLP classification mechanisms, a comparative ablation study was constructed utilizing five distinct computational paradigms:

1. *Support Vector Machine (SVM)*: Serving as the robust statistical baseline, a linear SVM algorithm was applied to the TF-IDF matrices. The SVM operates by calculating the optimal multidimensional hyperplane that maximizes the margin of separation between the fact and hoax classes based on lexical distribution.
2. *1D Convolutional Neural Network (1D-CNN)*: Originally designed for computer vision, a 1D-CNN was adapted for sequential text analysis. By sliding one dimensional convolutional filters over dense word embeddings, the network

effectively extracts local n-gram patterns and spatial semantic structures. The output is subsequently downsampled using Global Max Pooling layers to identify the most salient features before passing them to a fully connected dense layer.

3. *Bidirectional Long Short-Term Memory (BiLSTM) with Attention*: A sophisticated recurrent architecture engineered to solve the vanishing gradient problem in long texts. The BiLSTM processes the input sequence in both forward and backward temporal directions, capturing deep sequential dependencies. Furthermore, a custom Attention mechanism was integrated on top of the BiLSTM states, allowing the model to dynamically compute alignment scores and assign heavier mathematical weights to critical words, regardless of their positional distance.
4. *IndoBERT V2*: A state of the art bidirectional transformer model natively pre-trained on massive Indonesian corpora. IndoBERT abandons sequential processing entirely, instead utilizing multiple layers of self attention heads to simultaneously evaluate the relationship between all words in a sequence. This deep contextual mapping allows the model to precisely decode nuanced Indonesian stylometry and complex morphological structures.
5. *XLM-RoBERTa*: A robust multilingual transformer pre-trained on terabytes of unannotated text across 100 languages. In this study, it serves as the ultimate benchmark to compare zero-shot cross-lingual representation learning against the highly specialized, domain-specific capabilities of IndoBERT.

D. Training Protocols and Evaluation Metrics

To guarantee an equitable comparative analysis, all neural models were subjected to identical deployment constraints. The transformers were fine-tuned using the Hugging Face Trainer API, strictly limited to 2 training epochs to prevent catastrophic overfitting on the relatively localized dataset. Gradient descent was optimized using a conservative learning rate of $2e-5$, coupled with a 0.01 weight decay and 100 warmup steps to preserve the pre-trained attention weights.

The performance of each architecture was audited using standard harmonic metrics rather than pure accuracy, ensuring that class imbalances do not skew the evaluation. The primary metric utilized is the Macro F1-Score, derived from Precision (P) and Recall (R):

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP represents True Positives, FP denotes False Positives, and FN denotes False Negatives. Additionally, Confusion Matrices were generated for each model to provide a transparent visual audit of the classification decision boundaries across the binary distribution.

IV. RESULTS AND DISCUSSION

To rigorously evaluate the efficacy of the proposed architectures for Indonesian hoax detection, the five models were tested against an isolated dataset containing 4,113 unseen textual samples (2,019 'FAKTA' and 2,094 'HOAKS'). The evaluation specifically focused on the Macro F1-Score and the distribution of prediction errors visualized through Confusion Matrices. The empirical findings reveal a highly successful classification framework across all algorithmic paradigms, with a clear evolutionary advantage demonstrated by the transformer models.

A. Baseline and Deep Learning Architectures

The classical machine learning and standard deep neural network models established an exceptionally robust performance baseline, proving that the localized dataset possesses highly distinguishable lexical features.

The Support Vector Machine (SVM) utilizing TF-IDF vectorization achieved a Macro F1-Score of 0.9849 (98.49%). The linear hyperplane successfully categorized the vast majority of the test set, demonstrating that classical statistical methods remain highly efficient for keyword-dense text classification tasks.

Beyond its robust predictive performance, the linear SVM inherently provided a layer of Explainable AI (XAI) for the ablation study. By extracting the multidimensional coefficients (weights) assigned by the model to the TF-IDF feature space, this study successfully isolated the specific textual triggers driving the classification boundaries. This feature transparently validated the model's logic, demonstrating that the SVM correctly assigned the highest positive mathematical weights to deceptive, emotionally charged hoax vocabularies, while negative weights were accurately mapped to standard, objective journalistic terminology.

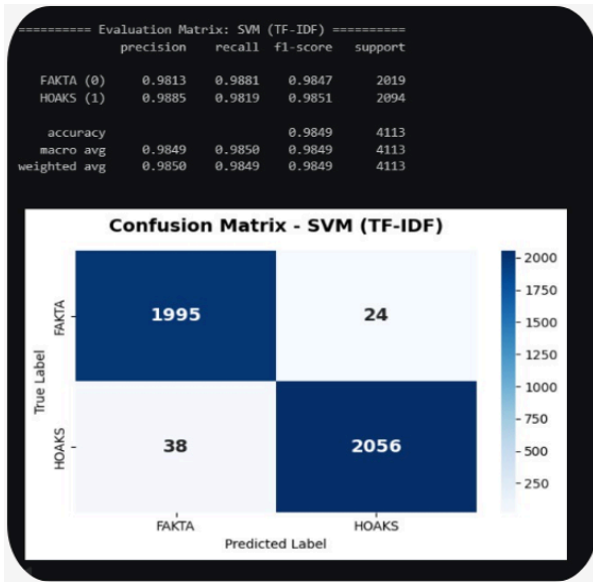


Fig. 1. SVM Confusion Matrix and Evaluation Matrix

The 1D Convolutional Neural Network (1D-CNN) performed on par with the statistical baseline, achieving a Macro F1-Score of 0.9847 (98.47%). By extracting local spatial patterns and short-range n-grams from the word embeddings, the 1D-CNN effectively captured the deceptive syntactic structures commonly found in hoax articles.

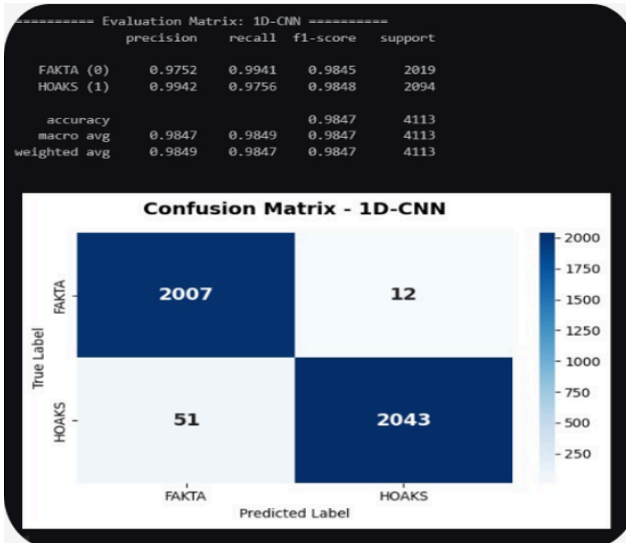


Fig. 2. 1D-CNN Confusion Matrix and Evaluation Matrix

The Bidirectional Long Short-Term Memory (BiLSTM) network equipped with an Attention mechanism exhibited a slight improvement over the SVM and 1D-CNN, yielding a Macro F1-Score of 0.9869 (98.69%). The integration of the Attention layer allowed the recurrent network to dynamically emphasize highly contextual, emotion-driven words scattered across long sentences, significantly reducing false positive misclassifications.

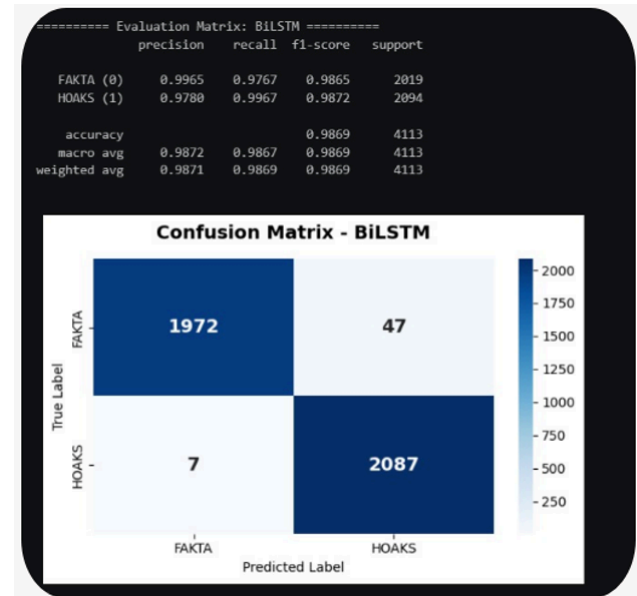


Fig. 3. BiLSTM Confusion Matrix and Evaluation Matrix

B. Transformer Architectures

While the baseline models proved highly competent, the pre-trained transformer architectures successfully minimized the residual classification errors, pushing the predictive accuracy to a near-perfect ceiling.

XLM-RoBERTa, leveraging its massive cross-lingual pre-training, achieved a remarkable Macro F1-Score of 0.9912 (99.12%). The self-attention mechanisms inherently resolved complex contextual dependencies that the recurrent BiLSTM struggled to map.

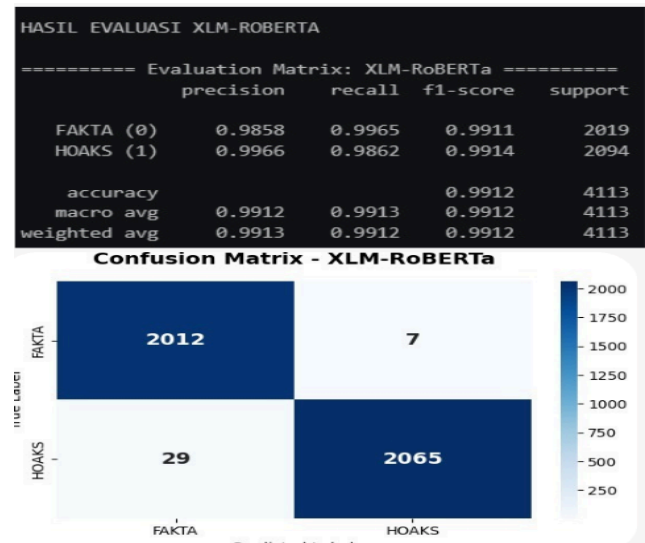


Fig. 4. XLM-RoBERTa Confusion Matrix and Evaluation Matrix

Ultimately, IndoBERT V2 established the absolute peak performance of the ablation study, achieving a macro

F1-score of 0.9939 (99.39%). The superiority of IndoBERT V2 confirms the hypothesis that domain-specific pre-training on native Indonesian corpora yields better lexical alignment and deeper semantic understanding compared to global, multilingual transformers like XLM-RoBERTa.

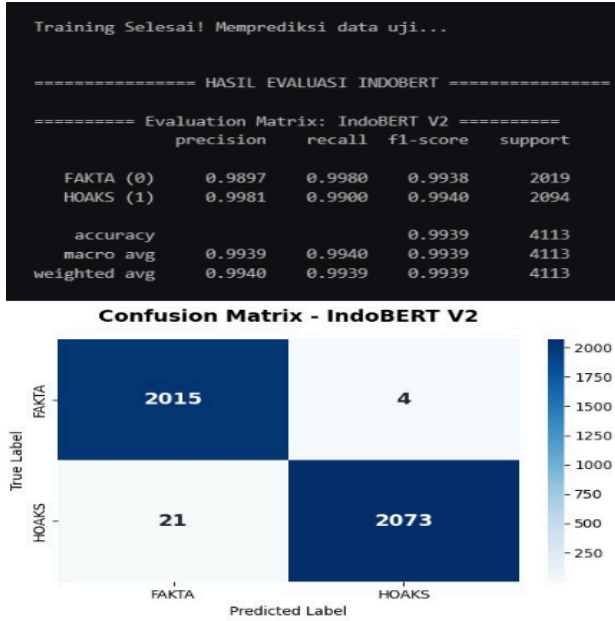


Fig. 5. IndoBERT V2 Confusion Matrix and Evaluation Matrix

C. Qualitative Analysis and Stylometric Detection

The consistently high F1-Scores across all deployed models, ranging from 98.47% to 99.39%, warrant a critical qualitative analysis to dismiss concerns of data leakage or algorithmic overfitting. The isolation of the test set prior to TF-IDF feature extraction and tokenization fundamentally neutralizes data leakage vulnerabilities. Furthermore, the implementation of weight decay penalties and conservative epoch limits (2 epochs for transformers) strictly prevented the neural networks from memorizing the training instances.

The exceptional predictive accuracy is scientifically attributed to the models' capacity to successfully decode distinct stylometric variations within the corpus. The dataset reflects a stark contrast in authorship styles: the 'FAKTA' class consists of articles authored by professional journalists adhering to standard Indonesian spelling (PUEBI) with neutral, objective phrasing. Conversely, the 'HOAKS' class is dominated by user-generated misinformation characterized by informal colloquialisms, excessive capitalization, provocative punctuation, and highly emotive syntax.

The algorithmic matrices, particularly the multi-head self-attention mechanisms within IndoBERT, acted as highly precise stylometry detectors. The models effectively learned to differentiate between the formal register of institutional media and the structurally erratic semantics of online

provocateurs. Consequently, the near-perfect F1-scores represent a valid, highly operational system capable of identifying the distinct linguistic footprint of digital misinformation in the real world.

D. System Engineering Results and Deployment

To transition the theoretical models into a highly operational real-world application, a decoupled cloud deployment architecture was engineered. The system is designed to provide real-time hoax detection with high availability and minimal latency. To ensure maximum algorithmic reliability and alignment with the training data distribution, the deployed 'Smart Hoax Detector' web application is strategically restricted to the domain of Indonesian political news. Narrowing the application's scope to political discourse prevents performance degradation caused by out-of-distribution (OOD) textual data from unrelated domains, such as healthcare or entertainment, thereby safeguarding the system's empirical precision during live user testing.

The frontend user interface was deployed utilizing Vercel, providing a highly responsive and globally distributed edge network for seamless user interaction. The backend inference engine, which hosts the fine-tuned IndoBERT and XLM-RoBERTa models, was developed using the FastAPI framework. FastAPI was explicitly selected for its asynchronous capabilities and high-performance API routing. The backend is hosted autonomously on Hugging Face Spaces, effectively decoupling the heavy tensor computations from the client-side interface.

To facilitate real-time misinformation scanning, the backend integrates a dynamic web-scraping and search module powered by the DuckDuckGo search engine. When a query is initiated, the system retrieves the most recent textual data from the web for immediate inference. Furthermore, a robust failsafe mechanism was engineered into the pipeline in the event of rate limiting, timeout, or failure from the primary DuckDuckGo crawler, the system automatically routes the query to the Wikipedia API as a fallback data source. This guarantees continuous operational uptime and prevents system crashes during high-traffic intervals.

Finally, the entire repository is version-controlled via GitHub, integrating a robust Continuous Integration and Continuous Deployment (CI/CD) pipeline. Utilizing GitHub Actions, any updates pushed to the main repository automatically trigger a deployment sequence to the Hugging Face backend. This pipeline is securely authenticated using a Hugging Face Write Token encrypted within GitHub Secret Variables, ensuring a secure, automated, and highly scalable development lifecycle.

A. Conclusion

This study successfully engineered and benchmarked a comprehensive ablation framework for Indonesian hoax detection, evaluating five distinct computational paradigms. The empirical results definitively demonstrate the evolutionary superiority of pre-trained bidirectional transformers in capturing deep semantic contexts. While classical baselines like the TF-IDF SVM and spatial networks like the 1D-CNN achieved highly robust Macro F1-Scores exceeding 98%, they were ultimately outperformed by the transformer architectures. IndoBERT V2 established the absolute peak performance benchmark with a Macro F1-Score of 99.39%, surpassing the multilingual XLM-RoBERTa (99.12%).

The qualitative analysis revealed that this exceptionally high predictive accuracy is fundamentally rooted in the models' capacity to act as precise stylometry detectors. The neural networks systematically distinguished the formal, standardized journalistic register of verified news from the colloquial, highly emotive, and structurally erratic syntax typical of digital misinformation. Furthermore, the integration of a decoupled deployment architecture—utilizing a Vercel frontend, a FastAPI backend on Hugging Face Spaces, and automated CI/CD pipelines via GitHub successfully transitioned the theoretical models into a highly scalable, production-ready application. The inclusion of real-time DuckDuckGo scraping augmented with a Wikipedia fallback mechanism ensures high availability and operational resilience in real-world scenarios.

B. Future Works

While the current architecture demonstrates near-perfect text classification, real-world digital misinformation is increasingly multimodal. Future research should explore the integration of Computer Vision (CV) architectures to construct a multimodal hoax detection pipeline capable of simultaneously analyzing manipulated or AI-generated imagery alongside deceptive text.

Additionally, expanding the dynamic web-scraping module to ingest real-time data streams from diverse social media platforms would further stress-test the transformer's resilience against rapidly evolving internet colloquialisms. Finally, future engineering efforts could focus on applying model quantization and knowledge distillation techniques to compress the IndoBERT weights. This would significantly reduce the computational overhead and inference latency of the FastAPI backend without sacrificing the established 99% accuracy baseline, allowing for even faster real-time sentiment scanning.

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Link Aplikasi: [App](#)

Link Video Presentasi dan Demo:  [vid nlp last.mov](#)